Problem Set 3

Applied Stats/Quant Methods 1
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Due: November 12, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 12, 2026. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in `R` using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

   R:

```r
fit1 = lm(formula = voteshare ~ difflog, data = inc.sub)
summary(fit1)
```

```
Call:
lm(formula = voteshare ~ difflog, data = inc.sub)
```

```
Residuals:
Min       1Q   Median       3Q      Max
-0.26832 -0.05345 -0.00377  0.04780  0.32749

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.579031   0.002251  257.19   <2e-16 ***
difflog     0.041666   0.000968   43.04   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.07867 on 3191 degrees of freedom
Multiple R-squared:  0.3673, Adjusted R-squared:  0.3671
F-statistic:  1853 on 1 and 3191 DF,  p-value: < 2.2e-16
```

On average, a one-unit increase in the difference in campaign spending (difflog) is associated with a 0.0417-unit increase in the incumbent's vote share. When there is no difference in campaign spending (difflog = 0), the predicted vote share for the incumbent is 0.5790 (57.9%).

2. Make a scatterplot of the two variables and add the regression line.

   R:

```r
library(ggplot2)
ggplot(data = inc.sub, aes(x = difflog, y = voteshare)) +
  geom_point() +
  labs(title = "Incumbent Vote Share vs. Spending Difference")
# add the regression line
ggplot(data = inc.sub, aes(x = difflog, y = voteshare)) +
  geom_point() +
  geom_smooth(method= "lm") +
  labs(title = "Incumbent Vote Share vs. Spending Difference")
```

   Figure 1 (scatterplot)

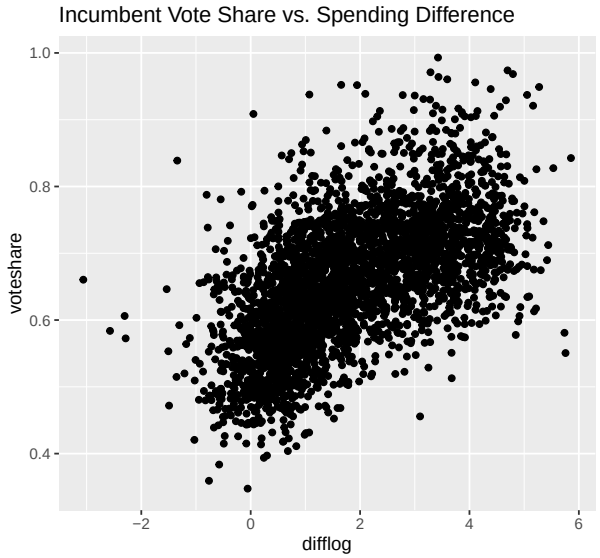
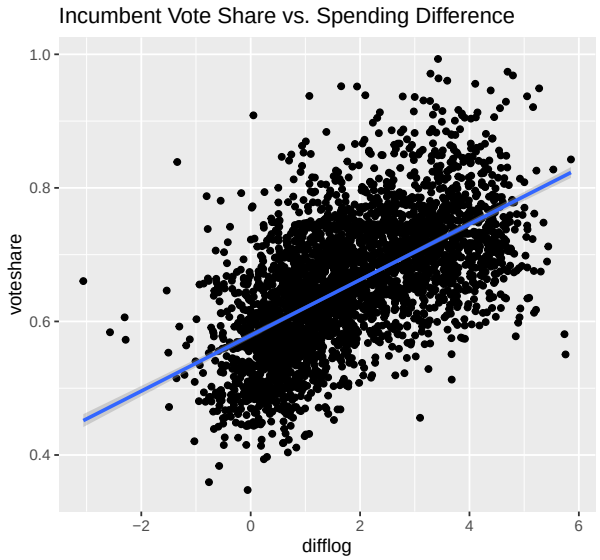


Figure 2 (scatterplot + regression line)



This scatterplot displays the positive relationship between the difference in campaign spending and the incumbent voteshare. The data points lie relatively close to the regression line, suggesting a strong linear relationship between the two variables.

3. Save the residuals of the model in a separate object.

   R:

```
res1 <- residuals(fit1)
head(res1)
```

```
            1             2             3             4             5
-0.0004227622 -0.0316840149 -0.0045514943  0.0386688767  0.0355287965  0.032283252
```

   The residuals represent the variation in the voteshare that is unexplained by the difflog (we remove the effect of difflog from the voteshare variable).

4. Write the prediction equation.

   $\widehat{\text{voteshare}} = 0.5790 + 0.0417 \times \text{difflog}$

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

   R:

```
fit2 = lm(formula = presvote ~ difflog, data = inc.sub)
summary(fit2)
```

```
Call:
lm(formula = presvote ~ difflog, data = inc.sub)

Residuals:
Min       1Q   Median       3Q      Max
-0.32196 -0.07407 -0.00102  0.07151  0.42743
```

```
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.507583   0.003161  160.60   <2e-16 ***
difflog     0.023837   0.001359   17.54   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.1104 on 3191 degrees of freedom
Multiple R-squared:  0.08795, Adjusted R-squared:  0.08767
F-statistic: 307.7 on 1 and 3191 DF,  p-value: < 2.2e-16
```

On average, a one-unit increase in the difference in campaign spending (difflog) is associated with a 0.0238-unit increase in the vote share of the presidential candidate of the incumbent's party. When there is no difference in campaign spending (difflog = 0), the predicted vote share for the presidential candidate of the incumbent's party is 0.5076 (50.8%).

2. Make a scatterplot of the two variables and add the regression line.

   R:

```r
ggplot(data = inc.sub, aes(x = difflog, y = presvote)) +
  geom_point() +
  labs(title = "Presidential Vote Share vs. Spending Difference")
# add the regression line
ggplot(data = inc.sub, aes(x = difflog, y = presvote)) +
  geom_point() +
  geom_smooth(method= "lm") +
  labs(title = "Presidential Vote Share vs. Spending Difference")
```

   Figure 3 (scatterplot)

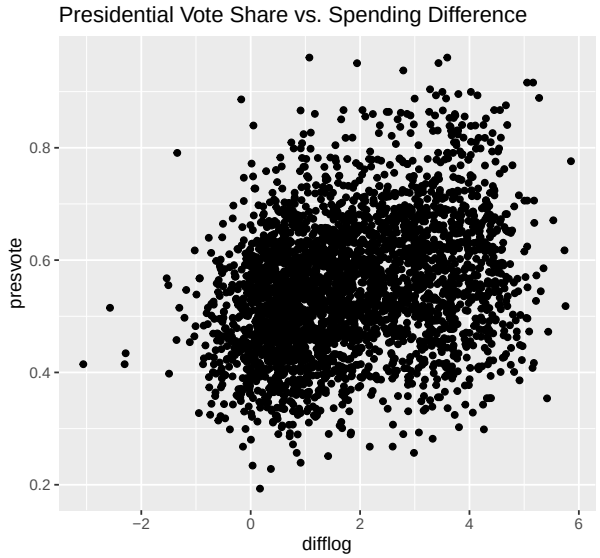
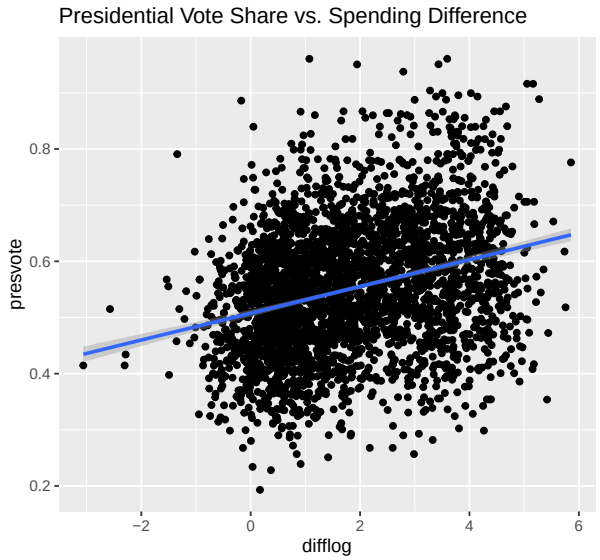


Figure 4 (scatterplot + regression line)



This scatterplot displays the positive relationship between the difference in campaign spending and the vote share of the presidential candidate of the incumbent's party. The data points show some dispersion around the regression line, suggesting a relatively weak model fit.

3. Save the residuals of the model in a separate object.

R:

```
res2 <- residuals(fit2)
head(res2)
```

```
           1            2            3            4            5            6
0.005605594  0.037578519 -0.053134788 -0.052993694 -0.045842994  0.074339701
```

The residuals represent the variation in the presidential voteshare that is unexplained by the difflog (we remove the effect of difflog from the voteshare variable).

4. Write the prediction equation.

$\widehat{\text{presvote}} = 0.5076 + 0.0238 \times \text{difflog}$

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

R:

```
fit3 = lm(formula = voteshare ~ presvote, data = inc.sub)
summary(fit3)
```

```
Call:
lm(formula = voteshare ~ presvote, data = inc.sub)

Residuals:
Min       1Q   Median       3Q      Max
-0.27330 -0.05888  0.00394  0.06148  0.41365

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.441330   0.007599   58.08   <2e-16 ***
presvote    0.388018   0.013493   28.76   <2e-16 ***
---
```

```
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.08815 on 3191 degrees of freedom
Multiple R-squared:  0.2058, Adjusted R-squared:  0.2056
F-statistic:   827 on 1 and 3191 DF,  p-value: < 2.2e-16
```

On average, a one-unit increase in the vote share of the presidential candidate of the incumbent's party (presvote) is associated with a 0.3880-unit increase in the incumbent voteshare. When the vote share of the presidential candidate of the incumbent's party is zero (presvote = 0), the predicted incumbent's vote share is 0.4413 (44.1%).

2. Make a scatterplot of the two variables and add the regression line.
   R:

```r
ggplot(data = inc.sub, aes(x = presvote, y = voteshare)) +
  geom_point() +
  labs(title = "Incumbent Vote vs. Presidential Vote Share")
# add the regression line
ggplot(data = inc.sub, aes(x = presvote, y = voteshare)) +
  geom_point() +
  geom_smooth(method= "lm") +
  labs(title = "Incumbent Vote vs. Presidential Vote Share")
```

Figure 5 (scatterplot)

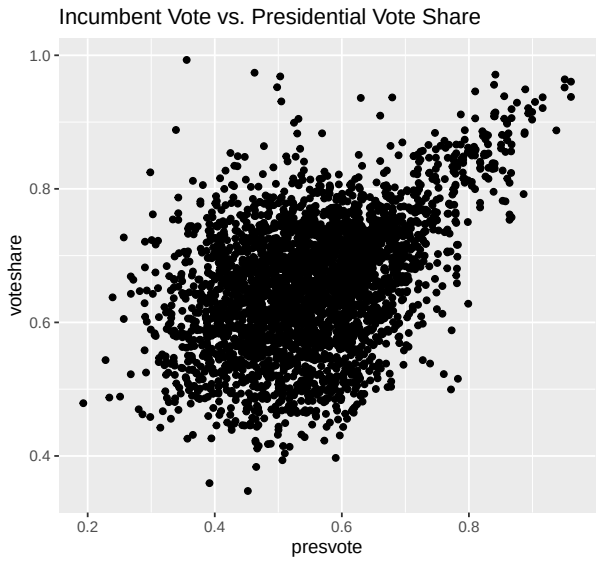
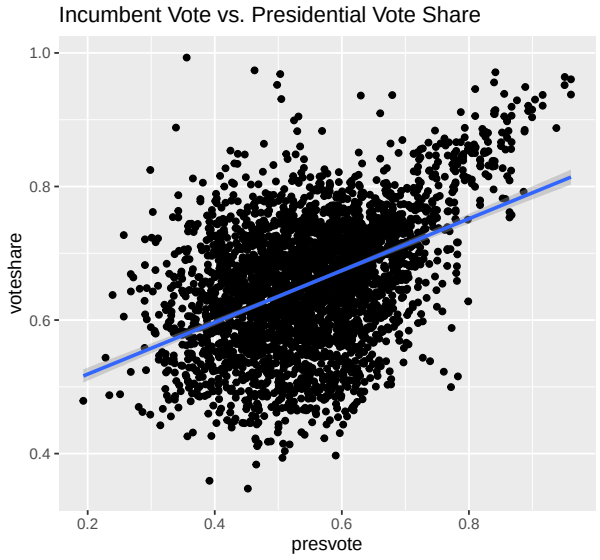


Figure 6 (scatterplot + regression line)



This scatterplot displays the positive relationship between the vote share of the presidential candidate of the incumbent's party and the incumbent voteshare. The data points show some dispersion around the regression line, suggesting a relatively weak model fit.

3. Write the prediction equation.

   $\widehat{\text{voteshare}} = 0.4413 + 0.3880 \times \text{presvote}$

Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

   R:

```
fit4 = lm(res1 ~ res2)
summary(fit4)
```

```
Call:
lm(formula = res1 ~ res2)

Residuals:
Min       1Q   Median       3Q      Max
-0.25928 -0.04737 -0.00121  0.04618  0.33126

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.942e-18  1.299e-03    0.00        1
res2         2.569e-01  1.176e-02   21.84   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07338 on 3191 degrees of freedom
Multiple R-squared:   0.13, Adjusted R-squared:  0.1298
F-statistic:   477 on 1 and 3191 DF,  p-value: < 2.2e-16
```

On average, a one-unit increase in the variation of voteshare unexplained by campaign spending difference (res2) is associated with a 0.2569-unit increase in the variation of presvote unexplained by campaign spending difference (res1). When the residual of voteshare unexplained by difflog is zero (res2 = 0), the predicted residual of presvote is approximately 0 (-1.942×10^{-18}).

2. Make a scatterplot of the two residuals and add the regression line.

   R:

```r
ggplot(data = inc.sub, aes(x = res2, y = res1)) +
  geom_point() +
  labs(title = "Residual of Question 1 vs. Residual of Question 2")
# add a regression line
ggplot(data = inc.sub, aes(x = res2, y = res1)) +
  geom_point() +
  geom_smooth(method= "lm") +
  labs(title = "Residual of Question 1 vs. Residual of Question 2")
```

   Figure 7 (scatterplot)

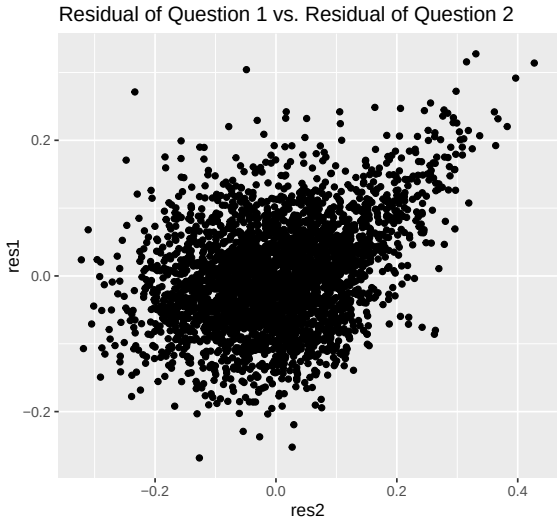
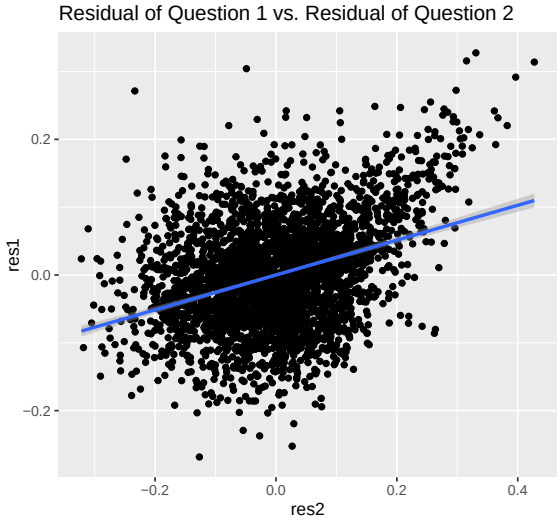


Figure 8 (scatterplot + regression line)



This scatterplot displays the positive relationship between the variation of voteshare unexplained by campaign spending difference (res2) and the variation of presvote unexplained by campaign spending difference (res1). The data points show some dispersion around the regression line, suggesting a relatively weak model fit.

3. Write the prediction equation.

   $\widehat{\text{res1}} = 0.2569 \times \text{res2}$

   The intercept is calculated as -1.942×10^{-18}, which is essentially zero.

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

   R:

```
fit5 = lm(formula = voteshare ~ difflog + presvote, data = inc.sub)
summary(fit5)
```

```
Call:
lm(formula = voteshare ~ difflog + presvote, data = inc.sub)

Residuals:
Min       1Q   Median       3Q      Max
-0.25928 -0.04737 -0.00121  0.04618  0.33126

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.4486442  0.0063297   70.88   <2e-16 ***
difflog     0.0355431  0.0009455   37.59   <2e-16 ***
presvote    0.2568770  0.0117637   21.84   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07339 on 3190 degrees of freedom
Multiple R-squared:  0.4496, Adjusted R-squared:  0.4493
F-statistic:  1303 on 2 and 3190 DF,  p-value: < 2.2e-16
```

   On average, a one-unit increase in the difference in campaign spending (difflog) is associated with a 0.0355-unit increase in the incumbent's vote share, holding the presidential vote share constant. In addition, on average, a one-unit increase in the presidential vote share (presvote) is associated with a 0.2569-unit increase in the incumbent's vote share, holding the difference in campaign spending constant. When there is no difference in campaign spending (difflog = 0) and the presidential vote share is 0%, the predicted incumbent's vote share is 0.4486 (44.9%).

2. Write the prediction equation.

   $\widehat{\text{voteshare}} = 0.4486 + 0.0355 \times \text{difflog} + 0.2569 \times \text{presvote}$

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case? The regression coefficient and SE of the two residuals in Q4

and the regression coefficient and SE of presvote in Q5 are exactly the same. In Q1, the residuals represent the part of voteshare unexplained by difflog (we remove the effect of difflog from the voteshare variable). Similarly, in Q2, the residuals represent the part of presvote unexplained by difflog (we remove the effect of difflog from the presvote variable). Therefore, in Q4, we regress the Q1 residuals on the Q2 residuals. This allows us to control for the effect of difflog by removing it from both variables, leaving only the effect of presvote on voteshare. This results in a coefficient of 0.2569 and an SE of 0.01176.

```
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.942e-18  1.299e-03    0.00        1
res2         2.569e-01  1.176e-02   21.84   <2e-16 ***
```

On the other hand, the multiple regression model in Q5 gives exactly the same results. Since multiple regression coefficients can statistically control for the effects of other variables in the model, the regression coefficient of presvote in Q5 represents its effect on voteshare independent of difflog. This results in the exact same coefficient of 0.2569 (0.2568770) and SE of 0.01176.

```
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.4486442  0.0063297   70.88   <2e-16 ***
difflog     0.0355431  0.0009455   37.59   <2e-16 ***
presvote    0.2568770  0.0117637   21.84   <2e-16 ***
```